

Shubham Bhattacharya

AI Agent Backend Engineer — TypeScript, Node.js, LangGraph.js

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SUMMARY

AI Agent Backend Engineer specializing in production LLM systems, multi-agent orchestration, and intelligent automation with TypeScript and Node.js. Built agentic backends using LangGraph.js state machines with deterministic routing, Zod-validated outputs, and human-in-the-loop patterns for financial safety. Strong focus on zero-trust AI architecture where every LLM output is untrusted until validated. Seeking AI Agent Backend Developer roles building production-grade agent systems.

TECHNICAL SKILLS

AI Agent Development: LangGraph.js, Multi-Agent Orchestration, Tool Calling, Human-in-the-Loop, Prompt Engineering

LLM Integration: Multi-Provider LLM Orchestration, LLM Output Validation, tiktoken, Streaming (SSE)

Retrieval & Embeddings: pgvector, RAG Pipelines, Semantic Search, BM25, Vector Databases

Languages: TypeScript, JavaScript, SQL

Backend & APIs: Node.js, Express.js, Next.js, REST APIs, SSE, Socket.IO, WebSockets

Database & Cache: PostgreSQL, Prisma, Redis, BullMQ, MongoDB

Validation & Auth: Zod, Clerk, JWT, bcrypt, Helmet

Observability: Langfuse, OpenTelemetry, Sentry, Pino

DevOps & Infra: Docker, GitHub Actions, Vercel

PROJECTS

Gigi AI Ops — Production AI Agent Backend

LangGraph.js, TypeScript, Zod, pgvector, PostgreSQL, BullMQ, Redis

- Architected multi-step AI agent backend for e-commerce operations using LangGraph.js state machines — deterministic intent classification, tool calling, and graceful failure recovery with automatic retries
- Built multi-provider LLM orchestration layer with automatic fallback across Groq, Gemini, and OpenRouter — per-persona routing with cost and latency tradeoffs
- Implemented hybrid RAG pipeline combining BM25 keyword search and dense vector search via pgvector for zero-hallucination FAQ retrieval with ranked results
- Enforced zero-crash reliability via Zod runtime validation on every LLM output — unstructured AI responses never reach the database or downstream systems
- Integrated tiktoken WASM for per-request token usage tracking and cost calculation in USD/INR with memory-safe cleanup
- Designed human-in-the-loop approval flow for financial operations — refunds above threshold require admin review, AI never moves money autonomously

github.com/shubhambhattacharya-dev/Agentic_Project | agentic-project-drab.vercel.app

Medo Copilot — Multi-Provider AI Launch Auditor

Next.js, TypeScript, Vercel AI SDK, Playwright, Clerk, Tailwind CSS

- Built AI-powered launch auditor with multi-provider LLM waterfall fallback: Gemini 2.0, Groq/Llama 4 Scout, OpenRouter/Claude — zero downtime on provider failures
- Designed hybrid scoring engine: 60% AI-generated insights + 40% deterministic metrics from Lighthouse performance audits and static code analysis
- Implemented Playwright-based visual intelligence capturing desktop and mobile screenshots for multimodal AI review of UI/UX quality
- Built static security analyzer scanning for hardcoded secrets, dependency vulnerabilities, and code quality issues with structured JSON output

github.com/shubhambhattacharya-dev/Medo_Copilot | medo-copilot.shubhambhattacharya.dev

DocNow — Telemedicine SaaS Platform

Next.js, TypeScript, Clerk, Prisma, PostgreSQL, Vonage, shadcn/ui

- Built full-stack telemedicine platform with WebRTC video calls via Vonage API, role-based dashboards for Patients, Doctors, and Admins
- Designed credit-based payment system with full audit trail: purchases, deductions, refunds, and earnings with transaction history
- Implemented doctor payout system with platform fee calculation, withdrawal requests, and admin approval workflow

github.com/shubhambhattacharya-dev/health | docnow-zeta.vercel.app

EDUCATION

BCA — Parul University

2022 — 2025 | CGPA: 8.0/10

Relevant Coursework: Data Structures & Algorithms, DBMS, Computer Networks, Operating Systems, Web Technologies